

Profile

Analytically driven MSc Business Analytics candidate (University of Nottingham, 2026) with a strong Computer Science foundation. Skilled in SQL, Python (scikit-learn, Pandas, NumPy), machine learning, and data visualisation, with hands-on project delivery across churn prediction, customer segmentation, and NLP-driven brand analytics. Proven ability to translate complex analysis into clear, actionable insights and communicate findings to stakeholders at all levels. Ready to bring data-driven thinking and partnership enthusiasm to a team where both genuinely matter.

Education

MSc Business Analytics, University of Nottingham, UK

Sep 2025 - Sep 2026

Modules: Foundational Business Analytics, Data at Scale: Management, Processing and Visualisation, Analytics Specialisations and Applications, Leading Big Data Business Projects, Machine Learning and Predictive Analytics, Supply Chain Planning and Management, and Advanced Operations Analysis.

Nottingham Advantage Award - Enterprise Mobility Business Challenge, University of Nottingham, UK

Nov 2025 - Mar 2026

- Collaborated in a team of 3 to develop a business case for launching a new Enterprise Car Club within a combined budget of £100,000, defining project objectives, benefits, and investment justification.
- Managed a £50,000 marketing budget, engaging stakeholders and defending decisions in a live Q&A pitch to Enterprise Mobility representatives - demonstrating benefit tracking and investment justification.
- Led location requirements analysis, comparing current-state constraints against target-state criteria to recommend optimal sites for accessibility and commercial viability.
- Developed and presented a launch strategy incorporating stakeholder needs, market research, and impact analysis for key business decisions.

Bachelor of Computer Applications, Maharaja Surajmal Institute, India

Aug 2022 - Aug 2025

Grade: CGPA: 8.27 (First class equivalent).

Key Modules: Python programming, Data structures and algorithm using C and C++, HTML, CSS, PHP, Java Script, and SQL.

Work Experience

Student Warden, Clarendon Street, Flow Student Accommodation, Nottingham, UK

Oct 2025 - Present

- Acted as first-line responder for **35+** out-of-hours incidents, conducting **7** welfare checks monthly and enforcing accommodation policies to de-escalate conflicts and enhance safety in a student residential building.
- Facilitated student wellbeing by providing pastoral guidance and signposting to university services, engaging with **100+** students to foster a positive residential community throughout the academic year.

Projects

Customer Churn Prediction (FoodCorp), University of Nottingham

Feb 2026 - Apr 2026

- Built a churn prediction pipeline on **26,454 labelled observations** across 18 reference dates; engineered **18 behavioural features** (recency, frequency, spend, basket diversity, trend).
- Evaluated 4 models (Logistic Regression, Random Forest, AdaBoost, SVM) against a **67.9% naive baseline**; selected a tuned Random Forest (200 trees, depth 6).
- Final test performance on **5,841 unseen customers**: ROC-AUC **88.1%**, Recall **83.4%**, Accuracy **78.6%** -AUC stable across all test dates (**0.876-0.888**).
- Analysed churn drivers, revealing visit frequency decline accounted for 28.3% of feature importance, and provided actionable recommendations to the product team, enhancing user retention strategy.

Social Media Brand Analysis: Ryanair vs. Wizz Air, University of Nottingham

Feb 2026 - Apr 2026

- Collected **26,265 raw posts** via BlueSky API; cleaned to **3,368 posts from 1,240 authors** after deduplication, bot filtering and aviation-tracker removal.
- Applied dual-engine sentiment analysis (TextBlob + VADER): Ryanair scored **+0.029** vs Wizz Air **+0.049**; Ryanair drove higher engagement (**4.94 avg likes vs. 1.42**).
- Ran **LDA topic modelling (k=4)** per brand, validated with **BERTopic on 1,000-post embeddings**; built a **500-feature TF-IDF perceptual map** (PCA-projected) for brand positioning.
- Segmented **1,240 authors** via KMeans (k=3) and built a **bipartite network (783 nodes, 102,737 edges)** to score micro-influencer candidates on a 5-dimension composite.
- Identified top influencers from **537 candidates**: @garius (score **0.446**, 29,784 followers) for Ryanair and @margareterosali (score **0.538** - highest in pool, 45,296 followers) for Wizz Air.

Retail Customer Segmentation for Data-Driven Marketing Strategy, University of Nottingham, UK

Jan 2026 - Mar 2026

- Interpreted transactional data of **3,000 retail customers** to recognize behavioural shopping patterns.
- Engineered **27 behavioural features** including **RFM** metrics, basket behaviour, and category spending.
- Applied **PCA** dimensionality reduction retaining 9 components explaining **~85% variance**.
- Implemented **K-Means clustering** to identify **5 customer segments**, generating operational insight to support Sales & Demand Planning and targeted marketing strategies.

Predictive Modelling for Marketing Campaign Response, University of Nottingham, UK

Dec 2025 - Jan 2026

- Analysed **4,000 customer** records from a bank marketing campaign dataset to predict product subscription behaviour, directly analogous to user sign-up conversion analysis.
- Conducted exploratory data analysis to identify key variables influencing campaign success.
- Built and evaluated **Logistic Regression, Decision Tree, kNN, and Random Forest models** for classification.
- Selected Random Forest as the best-performing model achieving **~0.85 accuracy** and highest F1-score (**0.49**) to support targeted marketing strategies and improve conversion rates for product sign-up funnels.

Retail Store Performance Analytics Dashboard, University of Nottingham, UK

Oct 2025 - Dec 2025

- Analysis of **retail transactional data** done across 4 stores to evaluate operational revenue performance and standardise reporting on customer spending behaviour.
- Developed analytical insights using KPIs including monthly revenue per store and average basket value (**£22-£59**).
- Evaluated revenue per active customer reaching up to **~£70**, identifying high-performing stores.
- Designed a Tableau dashboard tracking 4 standardised performance metrics across XX stores, maintaining a reliable data pipeline and exploring Power BI as an alternative reporting platform.

The Impacts of Social Media on Youth, Maharaja Surajmal Institute, India

Jan 2025 - Mar 2025

- Conducted survey of **600+ youths** (age 14-25), achieving **95%** response rate and comprehensive data for analysis.
- Applied statistical methods improving data accuracy by **20%** and created visualizations that enhanced insight clarity by **25%**.
- Authored comprehensive research paper with analytics-led graphical methods, aiding in academic and presented at **5th International Conference on ICT for Digital, Smart, and Sustainable Development**.

Extra-Curricular

Course Representative, MSc Business Analytics University of Nottingham, UK

Oct 2025 -Present

- Elected to represent the MSc Business Analytics cohort, attending Staff-Student Forum (SSF) meetings to channel student feedback directly to academic staff.
- Raised a course-related issue, resulting in departmental approval for implementation, enhancing the experience for future cohorts.

Cultural Head of the Institute, Maharaja Surajmal Institute, India

Sep 2024 - Jun 2025

- Spearheaded annual cultural festival for 2 years, managing a team of 15 students, resulting in 40% increase in student participation.
- Arranged the venue, food and drink, and entertainment, promoting the event on social media.

Certifications

Data Fundamentals, IBM SkillsBuild

Mar 2025

Getting Started with Data, IBM SkillsBuild

Mar 2025

Business Analysis and Process Management, Coursera

Feb 2024

Skills

IT: Python (NumPy, Pandas, Matplotlib, Seaborn, scikit-learn), SQL, Machine Learning (classification, clustering, NLP, feature engineering), Statistical Analysis, Data Visualisation (Tableau, Power BI), ETL concepts, Microsoft Office Suite (Excel, PowerPoint, Word, Teams), HTML, CSS.

Language: English (Fluent), Hindi (Native).